

Quadratic functions - Set 1 - Solutions

Specification MM1.3, M4.5, M4.11-M4.14.

Answer key

Question	Answer	Topic
1	E	Quadratic functions
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Worked solutions

Question 1 QUA-01 · Quadratic functions · spec MM1.3

The equation $x^2 + (k - 2)x + (k + 1) = 0$ has two distinct real roots. Find the complete set of possible values of k .

Answer: E

Discriminant $(k - 2)^2 - 4(k + 1) = k^2 - 8k = k(k - 8) > 0$, so $k < 0$ or $k > 8$.

Full method - how and why

'Two distinct real roots' is always a discriminant condition. The unknown is in the coefficients, so the discriminant becomes a quadratic in k and you finish with an inequality.

- 1 Identify $a = 1$, $b = k - 2$, $c = k + 1$ and write the discriminant: $(k - 2)^2 - 4(k + 1)$.
why: 'Two distinct real roots' $\Leftrightarrow b^2 - 4ac > 0$. Everything else in the question is decoration.
- 2 Expand and simplify: $k^2 - 4k + 4 - 4k - 4 = k^2 - 8k$.
why: The constants cancel, which is the sign you have expanded correctly.
- 3 Factorise and solve $k(k - 8) > 0$.
why: A quadratic inequality needs its roots and its shape: roots 0 and 8, opening upwards.
- 4 The upward parabola is positive outside its roots: $k < 0$ or $k > 8$.
why: Sketch it if in doubt - 'outside the roots' for > 0 , 'between the roots' for < 0 .

Where it goes wrong

- Expanding $(k - 2)^2$ as $k^2 - 4$, which gives $k^2 - 4k - 8$ and a different answer set.
- Reading the sign chart backwards and choosing $0 < k < 8$ - that is where the discriminant is **negative** and there are no real roots.

Takeaway Conditions on the number of roots are conditions on $b^2 - 4ac$; if the coefficients contain a parameter, you end up solving a quadratic inequality in that parameter.

Question 2 QUA-02 · Quadratic functions · spec MM1.3

By completing the square, find the minimum value of $2x^2 - 12x + 23$.

Answer: B

$2x^2 - 12x + 23 = 2(x - 3)^2 - 18 + 23 = 2(x - 3)^2 + 5$, so the minimum is 5 (at $x = 3$).

Full method - how and why

Minimum value of a quadratic: complete the square. The leading coefficient 2 must be factored out **before** completing the square.

- 1 Factor the 2 out of the x terms: $2(x^2 - 6x) + 23$.
why: Completing the square only works on $x^2 + (\text{something})x$; the coefficient of x^2 must be 1 inside the bracket.
- 2 Complete the square inside: $x^2 - 6x = (x - 3)^2 - 9$, so the expression is $2[(x - 3)^2 - 9] + 23$.
why: Halve the coefficient of x ($-6 \rightarrow -3$) and subtract the square of that half.
- 3 Multiply out the 2: $2(x - 3)^2 - 18 + 23 = 2(x - 3)^2 + 5$.
why: The -9 is inside the bracket, so it is multiplied by 2 as well - this is where marks are lost.
- 4 $2(x - 3)^2 \geq 0$, so the least value is 5, at $x = 3$.
why: A square is never negative, and it is zero exactly at the vertex.

Where it goes wrong

- Forgetting to multiply the -9 by 2, giving $2(x - 3)^2 + 14$ and the answer 14.
- Quoting the x -coordinate 3 instead of the minimum value 5 - both are offered.

Takeaway Take the leading coefficient outside first, complete the square, then multiply back in; the constant left outside is the minimum.

Question 3 QUA-03 · Quadratic functions · spec MM1.3

The quadratic $y = ax^2 + bx + c$ has its vertex at $(2, -3)$ and passes through $(0, 5)$. Find $a + b + c$.

Answer: A

$y = a(x - 2)^2 - 3$; at $x = 0$: $4a - 3 = 5$, so $a = 2$. Then $y = 2(x - 2)^2 - 3 = 2x^2 - 8x + 5$, and $a + b + c = 2 - 8 + 5 = -1$ (which is just $y(1)$).

Full method - how and why

Given a vertex, the completed-square form is the natural one. The clever final step is noticing that $a + b + c$ is just $y(1)$.

- 1 Write the completed-square form using the vertex: $y = a(x - 2)^2 - 3$.
why: The vertex (h, k) appears directly in $y = a(x - h)^2 + k$, so two of the three unknowns are already known.
- 2 Use the point $(0, 5)$: $a(0 - 2)^2 - 3 = 5 \Rightarrow 4a = 8 \Rightarrow a = 2$.
why: One extra point is exactly what is needed to pin down the remaining constant.
- 3 Note that $a + b + c$ is the value of $ax^2 + bx + c$ at $x = 1$.
why: Substituting $x = 1$ makes every power of x equal to 1 - a standard shortcut worth remembering.
- 4 Evaluate: $y(1) = 2(1 - 2)^2 - 3 = 2 - 3 = -1$.
why: No need to expand into $2x^2 - 8x + 5$ at all.

Where it goes wrong

- Expanding and then mis-collecting: $2(x - 2)^2 - 3 = 2x^2 - 8x + 5$, and slips in the -8 give 3 or 7.
- Using the vertex as $(-2, -3)$ - in $a(x - h)^2 + k$ the sign of h is already flipped.

Takeaway $f(1)$ is the sum of the coefficients; and a given vertex means you should start from $a(x - h)^2 + k$.

Question 4 QUA-05 · Quadratic functions · spec MM1.3

For what value of c does the line $y = 2x + c$ touch the parabola $y = x^2 - 4x + 7$?

Answer: C

Equate: $x^2 - 6x + (7 - c) = 0$. Tangency needs discriminant zero: $36 - 4(7 - c) = 0$, so $8 + 4c = 0$ and $c = -2$.

Full method - how and why

'Touches' means exactly one intersection, which after substitution means a repeated root: discriminant zero.

- 1 Set the two expressions equal: $2x + c = x^2 - 4x + 7$.
why: Intersections of $y = f(x)$ and $y = g(x)$ are the solutions of $f(x) = g(x)$.
- 2 Collect into a standard quadratic: $x^2 - 6x + (7 - c) = 0$.
why: Everything to one side, so the discriminant test can be applied.
- 3 Set the discriminant to zero: $(-6)^2 - 4(1)(7 - c) = 0$.
why: Touching means a repeated root, i.e. the line meets the curve once - exactly the $b^2 = 4ac$ case.
- 4 Solve: $36 - 28 + 4c = 0 \Rightarrow 4c = -8 \Rightarrow c = -2$.
why: Watch the double negative: $-4(7 - c) = -28 + 4c$.

Where it goes wrong

- Sign slip in $-4(7 - c)$ giving $c = 2$, which is offered.
- Using $b^2 - 4ac > 0$ (crossing) rather than $= 0$ (touching).

Takeaway Tangency between a line and a curve: substitute, then set the discriminant of the resulting quadratic to zero.

Question 5 QUA-07 · Quadratic functions · spec MM1.3

How many real solutions does the equation $x^4 - 5x^2 + 4 = 0$ have?

Answer: B

Let $u = x^2$: $u^2 - 5u + 4 = (u - 1)(u - 4) = 0$, so $x^2 = 1$ or $x^2 = 4$, giving $x = \pm 1, \pm 2$: four solutions.

Full method - how and why

A quartic that is quadratic in x^2 . The trap is counting the solutions of the substituted equation rather than of the original.

1 Substitute $u = x^2$: $u^2 - 5u + 4 = 0$.

why: Only even powers of x appear, which is what makes the substitution work.

2 Factorise: $(u - 1)(u - 4) = 0$, so $u = 1$ or $u = 4$.

why: Two values of u - but u is not the answer.

3 Undo the substitution: $x^2 = 1$ gives $x = \pm 1$; $x^2 = 4$ gives $x = \pm 2$.

why: Each **positive** value of u produces **two** values of x ; a negative value would produce none.

4 Count: four real solutions.

why: The question asks how many, not what they are.

Where it goes wrong

- Answering 2 because the quadratic in u has two roots.
- Forgetting the negative square roots and answering 2 again.

Takeaway After a substitution, always translate back: each positive value of x^2 gives two values of x , zero gives one, negative gives none.

Question 6 QUA-08 · Quadratic functions · spec MM1.3

The equation $kx^2 + 4x + k = 0$ has no real roots. Find the complete set of values of k .

Answer: C

Need $16 - 4k^2 < 0$, i.e. $k^2 > 4$, so $k < -2$ or $k > 2$. (If $k = 0$ the equation is linear with one root, so $k = 0$ is excluded anyway.)

Full method - how and why

'No real roots' is the discriminant < 0 . The extra subtlety is the coefficient k appearing in the x^2 term, so $k = 0$ needs a thought.

1 Write the discriminant: $4^2 - 4(k)(k) = 16 - 4k^2$.

why: Here $a = k$ and $c = k$ - both the first and last coefficients are the parameter.

2 Set it negative: $16 - 4k^2 < 0 \Rightarrow k^2 > 4$.

why: No real roots $\Leftrightarrow b^2 - 4ac < 0$.

3 Solve $k^2 > 4$: $k < -2$ or $k > 2$.

why: $k^2 > 4$ means $|k| > 2$, which is two separate intervals, not $-2 < k < 2$.

4 Check $k = 0$ separately: the equation becomes $4x = 0$, which has the root $x = 0$.

why: If $k = 0$ it is not a quadratic at all, so the discriminant test does not apply - but $k = 0$ is excluded by the answer anyway.

Where it goes wrong

- Solving $k^2 > 4$ as $-2 < k < 2$ - that is the solution of $k^2 < 4$.
- Ignoring the possibility $k = 0$ entirely (harmless here, fatal in questions where 0 lies inside the answer set).

Takeaway $k^2 > c$ gives two outer intervals; and whenever the leading coefficient contains the parameter, check the value that makes it zero.

Question 7 QUA-10 · Quadratic functions · spec MM1.3

The quadratic $x^2 + px + q$ has minimum value -4 and one of its roots is 0. Which of the following could be the quadratic?

Answer: D

Minimum -4 means $x^2 + px + q = (x - h)^2 - 4$, with roots $h \pm 2$. One root is 0, so $h = \pm 2$: the quadratic is $x^2 - 4x$ or $x^2 + 4x$. Only $x^2 - 4x$ is offered.

Full method - how and why

Two pieces of information about a quadratic - its minimum and one root - pin the whole thing down. Working in completed-square form makes the roots visible.

- 1 Minimum -4 and leading coefficient 1 means $x^2 + px + q = (x - h)^2 - 4$ for some h .
why: A monic quadratic with minimum m must be $(x - h)^2 + m$; only the position h is unknown.
- 2 Its roots satisfy $(x - h)^2 = 4$, i.e. $x = h \pm 2$.
why: Solve the completed-square form directly rather than expanding.
- 3 One root is 0, so $h + 2 = 0$ or $h - 2 = 0$: $h = -2$ or $h = 2$.
why: Both cases are legitimate; the options decide which one is wanted.
- 4 That gives $x^2 + 4x$ or $x^2 - 4x$; the list offers $x^2 - 4x$.
why: Check it: $x^2 - 4x = (x - 2)^2 - 4$, minimum -4 , roots 0 and 4.

Where it goes wrong

- Picking $x^2 - 2x - 3$: its minimum is -4 but its roots are -1 and 3 , so 0 is not a root.
- Checking only the minimum (several options have minimum -4) or only the roots.

Takeaway Test every stated condition - an option that satisfies one of them is designed to catch you.

Question 8 QUA-11 · Quadratic functions · spec MM1.3

The curve $y = x^2 - 2ax + a^2 + a - 6$ has its vertex on the x -axis. Find the value of a .

Answer: B

Complete the square: $y = (x - a)^2 + a - 6$. The vertex is $(a, a - 6)$, which lies on the x -axis when $a = 6$.

Full method - how and why

'Vertex on the x -axis' means the minimum value is zero, i.e. the quadratic has a repeated root. Completing the square is quicker than the discriminant here.

- 1 Complete the square in x , treating a as a constant: $x^2 - 2ax + a^2 = (x - a)^2$, so $y = (x - a)^2 + (a - 6)$.
why: The first three terms are already a perfect square - that is why the question is built with a^2 in it.
- 2 Read off the vertex: $(a, a - 6)$.
why: In $(x - h)^2 + k$ the vertex is (h, k) .
- 3 On the x -axis means the y -coordinate is zero: $a - 6 = 0$, so $a = 6$.
why: 'On the x -axis' is a statement about y , not x - the commonest misread in this question.

Where it goes wrong

- Setting the x -coordinate to zero and answering $a = 0$.
- Using the discriminant but mis-expanding $4a^2 - 4(a^2 + a - 6)$, which should collapse to $-4(a - 6)$.

Takeaway Vertex on the x -axis $\Leftrightarrow$ repeated root $\Leftrightarrow$ discriminant zero - three descriptions of the same situation.

Question 9 QUA-12 · Quadratic functions · spec MM1.3

The roots of $2x^2 + bx + 18 = 0$ are equal. Find the sum of the possible values of b .

Answer: C

$$b^2 - 144 = 0, \text{ so } b = \pm 12; \text{ their sum is } 0.$$

Full method - how and why

'Equal roots' is the discriminant-zero condition; the twist is that it gives two values of b and the question wants their sum.

- 1 Write the discriminant with $a = 2$, $c = 18$: $b^2 - 4(2)(18) = b^2 - 144$.

why: Substitute carefully - $4ac$ is 144, not 36.

- 2 Set it to zero: $b^2 = 144$, so $b = 12$ or $b = -12$.

why: Equal (repeated) roots means the discriminant vanishes; a square equal to a positive number always has two solutions.

- 3 Add them: $12 + (-12) = 0$.

why: The two values are symmetric about zero, so the sum is 0 without any further work.

Where it goes wrong

- Giving only the positive root and answering 12.
- Reading 'sum of the possible values of b ' as 'the value of b '.

Takeaway $b^2 = k$ always has two solutions $\pm\sqrt{k}$; read whether the question wants one of them, both, or a combination.

Question 10 QUA-13 · Quadratic functions · spec MM1.3

Which of the following quadratics has its vertex in the second quadrant (where $x < 0$ and $y > 0$)?

Answer: D

Complete the square: A vertex $(-2, -3)$; B $(2, 1)$; C $(-1, -2)$; D $(-3, 1)$, which has $x < 0$ and $y > 0$; E $(1, 2)$. So D.

Full method - how and why

A 'which one' question about vertices. Complete the square on each option - but use the sign of the x -coordinate to eliminate most of them first.

- 1 Recall that for $y = x^2 + bx + c$ the vertex is at $x = -\frac{b}{2}$.

why: You only need the vertex position, not the whole completed square, to test the first condition $x < 0$.

- 2 Second quadrant needs $x < 0$, so $-\frac{b}{2} < 0$, i.e. $b > 0$: only $y = x^2 + 4x + 1$ and $y = x^2 + 6x + 10$ survive among the upward parabolas.

why: Eliminating on one condition first halves the work.

- 3 Compute the y -coordinates: $x^2 + 4x + 1 = (x + 2)^2 - 3$, vertex $(-2, -3)$: fails $y > 0$. $x^2 + 6x + 10 = (x + 3)^2 + 1$, vertex $(-3, 1)$: passes.

why: Now apply the second condition to the short list.

- 4 Check the downward ones quickly: $-x^2 - 2x - 3$ has vertex $(-1, -2)$ and $-x^2 + 2x + 1$ has vertex $(1, 2)$ - neither is in the second quadrant.

why: Worth 10 seconds of confirmation on a 'which one' question.

Where it goes wrong

- Reading the vertex of $(x + 3)^2 + 1$ as $(3, 1)$ - the sign flips.
- Mixing up the quadrants: the second quadrant is up and to the left.

Takeaway On 'which of these' questions, test the cheapest condition first and eliminate before doing any heavy algebra.

Question 11 QUA-15 · Quadratic functions · spec MM1.3

The graph of $y = x^2 + bx + c$ crosses the x -axis at $(1, 0)$ and $(5, 0)$. What is the y -coordinate of the vertex?

Answer: C

$$y = (x - 1)(x - 5), \text{ vertex at } x = 3: y = (2)(-2) = -4.$$

Full method - how and why

Given the two roots, use the factorised form and the symmetry of the parabola - no expansion needed.

- 1 Write the factorised form: $y = (x - 1)(x - 5)$.

why: Roots at 1 and 5 with leading coefficient 1 (given, since the equation is $x^2 + bx + c$).

- 2 The vertex lies midway between the roots: $x = \frac{1+5}{2} = 3$.

why: A parabola is symmetric, so its turning point sits on the axis of symmetry, exactly halfway between the roots.

- 3 Evaluate there: $y = (3 - 1)(3 - 5) = 2 \times (-2) = -4$.

why: Substituting into the factorised form is far quicker than expanding to $x^2 - 6x + 5$.

Where it goes wrong

- Averaging the roots but then substituting into the wrong expression, e.g. getting -9 from $(3)^2 - 6(3) + 5$ miscomputed.
- Giving the x -coordinate 3 instead of the y -coordinate.

Takeaway Midpoint of the roots gives the axis of symmetry; substitute into the factorised form for the vertex height.

Question 12 QUA-16 · Quadratic functions · spec MM1.3

Given that $x^2 + y^2 = 10$ and $x + y = 4$, find the value of xy .

Answer: C

$$(x + y)^2 = x^2 + y^2 + 2xy, \text{ so } 16 = 10 + 2xy \text{ and } xy = 3.$$

Full method - how and why

A symmetric pair of equations. Expanding $(x + y)^2$ links the two given quantities to xy directly.

- 1 Write the identity $(x + y)^2 = x^2 + y^2 + 2xy$.

why: This is the only identity that connects a sum, a sum of squares and a product.

- 2 Substitute the given values: $4^2 = 10 + 2xy$.

why: $x + y = 4$ and $x^2 + y^2 = 10$ are both given, so only xy is unknown.

- 3 Solve: $16 - 10 = 2xy \Rightarrow xy = 3$.

why: Halve at the end - forgetting to divide by 2 gives 6, which is offered.

Where it goes wrong

- Answering 6 by omitting the factor 2 in $2xy$.
- Trying to solve for x and y individually (they are 1 and 3) - slower and unnecessary.

Takeaway $(x + y)^2 = x^2 + y^2 + 2xy$ converts between the three symmetric quantities; recognise which two you are given.